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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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1st Edition • 2026

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This book is currently in demo mode.

Only one initial sample chapter is shown so you can start from a clean base.

1. HOLA MUNDO

Hola Mundo.

This is an initial sample chapter for TeachBook.

TeachBook is a template to create teaching books from Markdown files and notebooks. From one source, you can build a navigable website and, when needed, a PDF version.

2. THE TRANSFORMERS PAPER

This chapter summarizes the paper located in the project's recursos folder (`recursos/paper_transformers.pdf`): **Attention Is All You Need** (Vaswani et al., NeurIPS 2017).

2.1 Main idea

The core proposal is to replace recurrent and convolutional sequence models with an architecture based only on attention. That architecture is the **Transformer**.

Instead of processing tokens one by one like an RNN, the Transformer can process all tokens in parallel during training.

2.2 What problem it addressed

In machine translation and other sequence tasks, previous models:

- were slower to train due to sequential computation,
- had more difficulty modeling long-range dependencies,
- and scaled worse with sequence length.

The paper shows that self-attention improves both quality and training efficiency.

2.3 Key Transformer components

1. **Self-Attention**: each token attends to other tokens in the same sequence.
2. **Scaled Dot-Product Attention**: attention scores are scaled by $\sqrt{d_k}$.
3. **Multi-Head Attention**: multiple parallel heads capture different relations.
4. **Positional Encoding**: since there is no recurrence, position is injected with sinusoidal signals.
5. **Encoder-Decoder stacks**: attention + feed-forward blocks with residual connections and layer normalization.

2.4 Reported results

According to the paper:

- it outperformed prior state of the art on WMT14 English-German,
- achieved very strong performance on WMT14 English-French,
- and reduced training cost compared with previous approaches.

2.5 Why it was influential

This paper reshaped deep learning for language and later many other domains. Models such as BERT, GPT, and modern LLMs are built on Transformer foundations.

2.6 One-line takeaway

The paper shows that **attention alone** can be the core of sequence models that are more accurate, more parallelizable, and more scalable than classic recurrent alternatives.